

EDA

Exploratory

Your task is to understand the basic structure of a new data set from 2020 related to tuition costs at colleges and universities across the country. It is stored as a csv on github that can be read into your R session (or a Quarto document) using:

```
tuition_cost <- readr::read_csv('https://raw.githubusercontent.com/rfordatascience/tidytuesday/main/data/2020/2020-03-10/tuition_cost.csv')
```

For each of the following, please answer the question in full sentences, provide the code that you used, along with the plot (or a sketch of it).

1. What are the dimensions of the data set? What is the unit of observation? What is the nature of the variables (what they are, their units or levels, and their type)?
2. Is there any missing data? If so, where?
3. What is the distribution of the cost of in-state tuition? Answer with summary statistics, a plot, and a written description of its shape.

4. The `type` variable is intended to sort institutions into three types: public, private, and for-profit. Evaluate whether that is true of this data set and let it inform your answer to the next question.
5. How does the distribution of in-state tuition differ between the types of institutions? Answer with a plot and a written description of how the distributions compare.
6. Which are the three most expensive institutions? The three least expensive institutions?

7. Create one more visualization of at least two variables that you're curious about. Describe the structure that you see with a written description. Then write down one new question that your plot raises, one that you would need more digging (or more data) to answer.

Explanatory

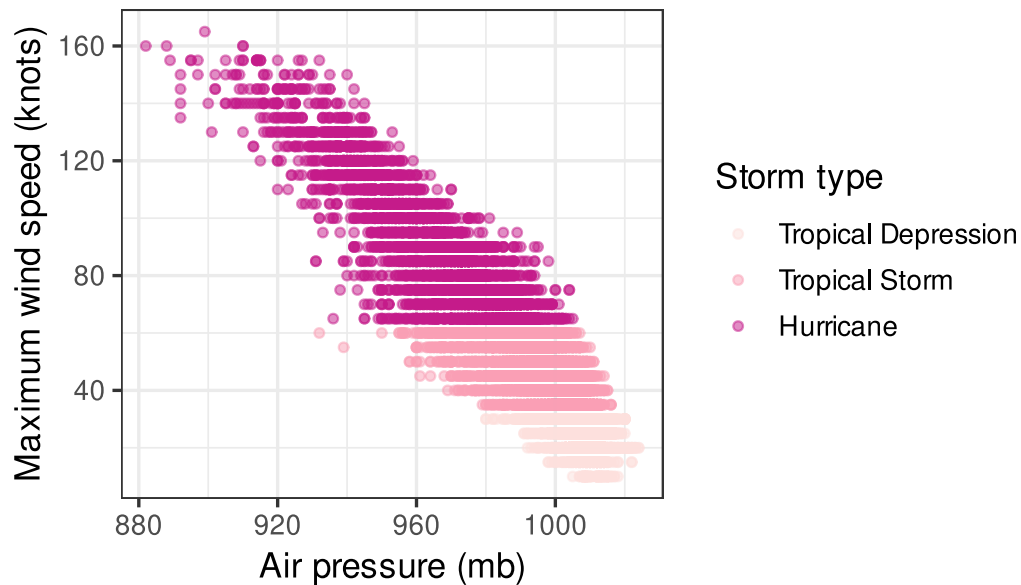
Each of the plots below is meant to tell a particular story to a particular audience. As you craft them, think carefully about the labels, themes, aesthetic settings, and the manner in which they facilitate comparisons.

8. Return to the tuition data and make two explanatory plots from it, each telling a different story. For each plot, before you write any code, write down a plan like the ones we made for the EV data in class:
 - **Audience:** who is this plot for?
 - **Story:** what is the one thing you want them to take away?
 - **Strategy:** what will you do to the plot to make that story easy to see?

Then provide the code alongside the plot (or a sketch of it). The two plots should make different comparisons easy, so draw on the tools from class: labels, limits and baselines, non-linear scales, reordering categories with `fct_reorder()` or `fct_infreq()`, themes, and setting aesthetics.

9. Provide the code that will create this plot from the `storms` data set inside the `dplyr` package. A few notes:
- Each dot is a single *observation* of a storm, not a whole storm: storms are measured every six hours over their lifetimes.
 - You will need to create a new column indicating storm type based on wind speed. For that you'll need to learn how that determination is made and then add it to the data frame. The `case_when()` function is a helpful shortcut similar to `ifelse()`.
 - The color palette belongs to a set of palettes called ColorBrewer. You can read more about them here: https://ggplot2.tidyverse.org/reference/scale_brewer.html.

Wind Speed and Pressure in Storms (1975-2024)



11. Remake your storms plot using a very specific visual theme: the botanical paintings of the US Department of Agriculture. They have been wrapped up into a ggplot2 theme inside the `ggpomological` package: <https://github.com/gadenbuie/ggpomological>. Read about how to use the package and write below the additional layers that you added to your plot above (for a challenge, see if you can get the handwriting font to show up). If you also wanted to tweak your plot with `theme()`, where in your code would that layer need to go, and why?

12. The `present` data frame stores data from the US Census about the number of boys and girls born each year between 1940 and 2013. It is stored in the `stat20data` package, which can be installed from GitHub using the `devtools` package.

```
# install.packages("devtools")
# devtools::install_github("https://github.com/stat20/stat20data")
library(stat20data)
slice(present, 1:5)
```

Make two plots of the trend over time in the proportion of births that are girls, each with a horizontal line at a baseline of .5 (see `geom_hline()`).

- **Plot A:** put the annual variability into context by ensuring the viewer can see the full range of possible values that the proportion could take.
- **Plot B:** let the y-axis zoom in on the range of the data.

For each plot, write one sentence giving the story it tells and an audience it would suit.

13. (Challenge) Some of the most effective plots on the web don't come from R or Python, but from dedicated visualization libraries written in JavaScript. That is the case for the EV registration plot made by the Department of Energy (<https://afdc.energy.gov/data/10962>), which uses a library called [Highcharts](#).

While you could craft the plot directly using JavaScript, a member of the R community has written a package that allows you to make them directly from R: you write R code, which the package turns into JavaScript, which Highcharts then renders as a plot. Read how to use the package on its documentation site: <https://github.com/jbkunst/highcharter/>.

Your task is to recreate the EV Highcharts plot exactly using the data found in an excel spreadsheet on the DoE website. You can export the excel spreadsheet as a csv file, as we did in class, or you can experiment with the `readxl` package to read it into R directly.

More questions coming soon . . .